

# WHEN THE RED QUEEN STUMBLES: Temporal Asymmetry and the Collapse of Regulatory Coevolution

**Ignacio Adrián Lerer**

Independent Researcher

Buenos Aires, Argentina

adrian@lerer.com.ar

ORCID: 0009-0007-6378-9749

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## ABSTRACT

Regulatory systems exhibit Red Queen dynamics: regulators evolve rules, regulated entities evolve workarounds, prompting new rules. This coevolution assumes temporal symmetry — both parties adapt at comparable velocities. Computational acceleration has put this assumption under severe stress. When regulated entities adopt algorithmic decision-making (adaptation cycle: days) while regulators operate at legislative tempo (adaptation cycle: years), the Evolutionary Velocity Ratio (EVR) exceeds 100:1. This paper argues that  $EVR > 100$  triggers qualitative regime change: regulatory systems cease to coevolve and instead collapse into what I term "administrative theater" — formal governance structures that cannot affect outcomes they nominally regulate.

Drawing on evolutionary biology, game theory, and institutional economics, I formalize three coevolutionary regimes (symmetric, asymmetric, collapsed) and test predictions across tax law ( $EVR \approx 5$ ), financial regulation ( $EVR \approx 100$ ), and high-frequency trading ( $EVR > 1000$ ). Quantitative analysis reveals a systematic EVR-effectiveness relationship: jurisdictions with  $EVR > 100$  exhibit regulatory arbitrage rates 15–25 times higher than matched controls, independent of regulatory stringency or resources.

The dominant reform proposal — “computational law” using AI adjudication to match regulatory velocity — eliminates rather than preserves the human agency regulation exists to protect. The most technically sophisticated version of this proposal, the JudgeAI system developed by Kozlov (2024), explicitly designs for “absolute independence” by making human influence on decision-making impossible — which is precisely what destroys the conditions for governance. I propose mandatory velocity constraints (circuit breakers,

cooling periods, algorithmic speed limits) that restore coevolutionary symmetry without sacrificing innovation.

**Keywords:** regulatory coevolution, computational velocity, evolutionary game theory, Red Queen hypothesis, extended phenotype, high-frequency trading, administrative law, algorithmic governance, JudgeAI, Nash equilibrium

**JEL Codes:** K23, D82, C73, L51, O33

## I. INTRODUCTION

### A. The Puzzle

On May 6, 2010, U.S. equity markets experienced the “Flash Crash”: the Dow Jones plunged nearly 1,000 points in minutes before recovering. High-frequency trading (HFT) algorithms, executing millions of orders per second, had triggered cascading liquidity withdrawals. The Securities and Exchange Commission (SEC) investigation took five months to produce preliminary findings.

This temporal gap — algorithms operating in microseconds, regulators responding in months — exemplifies a broader phenomenon transforming law’s relationship to the conduct it regulates. The SEC did not lack resources, expertise, or authority. It lacked something more fundamental: **temporal compatibility** with the systems it attempted to govern.

Traditional regulatory theory assumes what might be called the **coevolution postulate**: regulators and regulated entities engage in ongoing adaptation, each responding to the other’s moves in a process analogous to biological coevolution. Tax authorities close loopholes, tax lawyers find new ones, authorities respond. Antitrust agencies block mergers, firms restructure deals, agencies adapt doctrine. This Red Queen dynamic — named for the character in *Through the Looking-Glass* who must run continuously to stay in place — has defined regulatory scholarship for decades.

But the coevolution postulate rests on an unstated assumption: **temporal symmetry**. Both parties must operate on comparable timescales for mutual adaptation to occur. When this symmetry breaks — when one party’s adaptation cycle accelerates by orders of magnitude — coevolution collapses.

### B. The Argument

This Article makes three contributions:

**First**, I formalize the concept of **Evolutionary Velocity Ratio (EVR)**: the quotient of regulatory adaptation time to regulated entity adaptation time. EVR provides a dimensionless measure of temporal (a)symmetry in governance relationships.

**Second**, I identify three coevolutionary regimes distinguished by EVR magnitude: a Symmetric Regime ( $EVR \approx 0.5-2$ ) where both parties adapt at comparable velocities; an Asymmetric Regime ( $EVR \approx 2-100$ ) where regulators lag but eventually catch up; and a Collapsed Regime ( $EVR > 100$ ) where coevolution breaks down entirely and arbitrage becomes structural rather than tactical.

**Third**, I offer empirical evidence suggesting that  $EVR > 100$  triggers qualitative transformation. Using comparative analysis across tax law ( $EVR \approx 5$ ), securities regulation ( $EVR \approx 50-100$ ), and HFT ( $EVR > 1000$ ), I find that collapsed-regime jurisdictions exhibit regulatory arbitrage rates 15-25 times higher than symmetric-regime controls, holding constant regulatory stringency, resources, and political factors. These results are consistent with the regime trichotomy hypothesis, though the causal mechanisms warrant further empirical investigation.

### C. The Stakes

The temporal collapse of regulatory coevolution has profound implications across three dimensions. Theoretically, it undermines foundations of administrative law: Chevron deference, State Farm reasoned decision-making, notice-and-comment procedures — all presume regulators can meaningfully assess and respond to regulated conduct. When  $EVR > 100$ , these procedures become ceremonial. Practically, it explains otherwise puzzling regulatory failures. Why does financial innovation consistently outpace regulation despite massive SEC resources? Why do platform companies operate in persistent legal gray zones despite active enforcement? Not because regulators are captured or incompetent, but because they operate in different temporal dimensions than the entities they regulate. Normatively, it reveals why the dominant reform proposal — “computational law” using autonomous AI adjudication — is evolutionarily suicidal. Accelerating regulators to computational velocity eliminates the human agents regulation exists to protect. The solution is not faster law but **slower environments**: mandatory deceleration through circuit breakers, cooling periods, and algorithmic speed limits.

### D. Roadmap

Part II develops the theoretical framework, formalizing EVR and deriving predictions about regime transitions. Part III presents empirical analysis across three regulatory domains. Part IV examines the extended phenotype implications. Part V critiques computational law

proposals — including a close reading of the most technically advanced implementation currently in operation — and develops the mandatory deceleration alternative. Part VI addresses objections. Part VII concludes.

## II. THEORETICAL FRAMEWORK

### A. The Red Queen in Biology and Economics

#### 1. *Van Valen's Hypothesis*

Leigh Van Valen's 1973 "Red Queen Hypothesis" explained a paleontological puzzle: why do species exhibit constant extinction probabilities regardless of age? His answer: **organisms face continuously evolving competitors and parasites**. Each lineage's adaptations create selection pressure on others, requiring perpetual evolution merely to maintain relative fitness.

The hypothesis takes its name from Lewis Carroll's *Through the Looking-Glass*:

"Now, here, you see, it takes all the running you can do, to keep in the same place.  
If you want to get somewhere else, you must run at least twice as fast as that!"

Empirically, Van Valen showed that extinction probabilities remained roughly constant within taxonomic groups across geological time — exactly what one would expect if fitness is relative and competitors are also evolving.

#### 2. *Extension to Economics and Law*

Economists rapidly recognized Red Queen dynamics in strategic contexts. Arthur (1989) demonstrated increasing returns and path dependence in technology adoption; Fudenberg and Tirole (1991) analyzed dynamic games with arms-race structures; Dixit and Nalebuff (1991) examined strategic commitment as an attempt to win Red Queen races.

Legal scholars applied the metaphor to regulation: Kahan (1997) explored social norms versus law as coevolving systems; Bamberger and Mulligan (2011) traced privacy law chasing technological change; Wu (2013) described the "cycle" of information empires and regulatory intervention.

But this literature uniformly assumes **temporal symmetry**: both parties run at comparable speeds. The question of what happens when this assumption fails receives virtually no formal treatment.

## B. Formalizing Evolutionary Velocity Ratio (EVR)

### 1. Definitions

**Definition 1 (Adaptation Cycle):** An adaptation cycle  $\tau$  is the mean time required to complete the sequence: detect environmental change  $\rightarrow$  design response  $\rightarrow$  implement response  $\rightarrow$  validate effectiveness.

For regulators:  $\tau_R = \tau_{\text{detect}} + \tau_{\text{design}} + \tau_{\text{implement}} + \tau_{\text{validate}}$

For regulated entities:  $\tau_E = \tau_{\text{detect}} + \tau_{\text{design}} + \tau_{\text{implement}} + \tau_{\text{validate}}$

**Definition 2 (Evolutionary Velocity Ratio):** The Evolutionary Velocity Ratio is  $\text{EVR} = \tau_R / \tau_E$

EVR is dimensionless (ratio of times).  $\text{EVR} = 1$  indicates symmetric coevolution;  $\text{EVR} > 1$  indicates regulators adapt slower than entities; EVR is context-dependent, varying by domain, jurisdiction, and technology.

### 2. Regime Classification

**Theorem 1 (Regime Trichotomy):** Regulatory systems exhibit three qualitatively distinct regimes.

**Regime I (Symmetric):  $\text{EVR} \in [0.5, 2]$**  Both parties adapt at comparable velocities. Classical Red Queen coevolution operates; mutual adaptation is fitness-relevant. Arbitrage is profitable but risky, since enforcement eventually catches violators. Examples: traditional tax law, environmental compliance, labor law.

**Regime II (Asymmetric):  $\text{EVR} \in (2, 100]$**  Regulators lag but gaps are finite. Regulated entities maintain the lead but must continuously monitor regulatory changes. Arbitrage is systematically profitable for sophisticated actors; enforcement targets laggards. Examples: securities regulation pre-HFT, telecommunications regulation.

**Regime III (Collapsed):  $\text{EVR} > 100$**  Regulators govern obsolete versions of systems they nominally regulate. Coevolution breaks down; regulated entities operate independently of regulatory actions. Arbitrage is risk-free for all actors; enforcement is ceremonial. Examples: high-frequency trading, platform terms-of-service, algorithmic content moderation.

The trichotomy follows from information-theoretic bounds. When  $\tau_R \gg \tau_E$ , regulators' information about entity behavior is systematically outdated. At  $\text{EVR} \approx 100$ , the regulator's mental model of the entity is approximately two orders of magnitude behind reality — the threshold where **predictive collapse** occurs.

### 3. Connection to Game Theory

EVR collapse connects to game-theoretic agency conditions (Lerer, 2025a). Game theory requires five conditions for genuine strategic interaction: separate players with distinct preferences; imperfect but bounded information asymmetry; bounded rationality creating strategic uncertainty; minimal symmetry in affecting outcomes; and **temporal compatibility** — comparable decision velocities.

When  $EVR > 100$ , condition 5 fails catastrophically, amplifying failures of conditions 2–4. Information becomes arbitrarily asymmetric; the faster party can simulate the slower party with high accuracy, eliminating strategic uncertainty; and the slower party's actions arrive too late to affect outcomes, making power asymmetry absolute. Interaction transforms from game (mutual strategic adaptation) to **administration** (faster party unilaterally optimizes, treating slower party as environmental constant).

#### C. Why $EVR > 100$ Triggers Collapse

Three mechanisms explain qualitative regime change at the  $EVR \approx 100$  threshold.

**Predictive Horizon Collapse:** Regulatory action at time  $t$  aims to influence entity behavior at time  $t + \tau_R$ . But if entity adapts  $N$  times during  $\tau_R$ , and  $N > 100$ , the entity's state when regulation takes effect bears no relationship to the entity's state when regulation was designed. Formally: let  $S(t)$  be entity state at time  $t$ . Regulator observes  $S(t_0)$ , designs intervention for  $S(t_0 + \tau_R)$ . But entity has adapted  $N = \tau_R/\tau_E$  times. If  $N > 100$  and adaptations are non-trivial, the distance  $|S(t_0 + \tau_R) - S(t_0)|$  is so large that intervention designed for  $S(t_0)$  has no predictable effect on  $S(t_0 + \tau_R)$ .

**Feedback Loop Severance:** Effective regulation requires feedback — regulatory action  $\rightarrow$  entity response  $\rightarrow$  regulatory adjustment — with period  $\tau_R + \tau_E$ . In Regime I ( $EVR \approx 1$ ), a 10-year regulatory period contains roughly 10 adaptation cycles and learning is feasible. In Regime III ( $EVR > 100$ ), a 10-year period contains fewer than 0.1 regulator cycles, making learning impossible within policy-relevant timeframes.

**Structural Arbitrage:** In Regimes I–II, regulatory arbitrage is tactical — entities exploit current gaps but face the risk that regulators will close them. In Regime III, arbitrage is structural: entities can predict with high confidence that any specific gap will persist longer than the commercial lifespan of exploiting it. Arbitrage is not “gaming the system” — it is standard operating procedure.

III. EMPIRICAL ANALYSIS

A. Measurement Strategy

Testing regime predictions requires measuring EVR across regulatory domains. For  $\tau_R$  (regulator adaptation), I use: legislative mean time from bill introduction to enactment plus implementation; administrative mean time from NPRM to final rule plus enforcement; and judicial mean time from case filing to precedent solidification. For  $\tau_E$  (entity adaptation), I use: compliance officer survey data on detection-to-implementation lag; patent data as proxy for innovation velocity; frequency of terms-of-service updates for platforms; and code commit frequency for algorithmic systems.

B. Case Study 1: Tax Law (Regime I — Symmetric)

**Context:** U.S. federal income tax law, 1990–2020.

**EVR Estimate:**  $\tau_R \approx 3\text{--}5$  years (IRS notice-and-comment plus enforcement ramp-up);  $\tau_E \approx 6\text{--}18$  months (tax planning innovation cycle); **EVR  $\approx 3\text{--}5$**  (Regime I, upper bound).

**Empirical Validation** using IRS Statistics of Income data and tax shelter prosecutions 1990–2020:

| Metric                                               | Prediction | Observed         |
|------------------------------------------------------|------------|------------------|
| Returns with aggressive positions caught within 10yr | 40–60%     | 47%              |
| Major loophole lifespan                              | 3–7 years  | 4.2 years (mean) |
| Settlement rate (audit vs. litigation)               | High       | 83%              |

Tax law exhibits characteristic Regime I dynamics. Sophisticated strategies — corporate inversions, COLI schemes — are eventually closed through legislation or IRS guidance. The coevolutionary relationship is asymmetric but functional.

C. Case Study 2: Securities Regulation (Regime II/III Transition)

**Context:** U.S. equity markets, 1995–2020, focusing on HFT adoption.

**EVR Evolution:**

Period 1 (1995–2005, pre-HFT):  $\tau_R \approx 18\text{--}24$  months;  $\tau_E \approx 6\text{--}12$  months; **EVR  $\approx 2\text{--}3$**  (Regime I/II boundary).

Period 2 (2006–2020, HFT era):  $\tau_R \approx 18\text{--}24$  months (unchanged);  $\tau_E \approx$  days to weeks; **EVR  $\approx 50\text{--}500$**  (Regime III).

| Metric                            | Pre-HFT<br>(2000–<br>2005) | HFT Era<br>(2010–<br>2015) | Ratio     |
|-----------------------------------|----------------------------|----------------------------|-----------|
| Volume in dark pools              | 3–5%                       | 18–40%                     | 6–8×      |
| Flash crash events per year       | 0.2                        | 4.5                        | 22×       |
| Avg SEC enforcement lag           | 14 months                  | 38 months                  | 2.7×      |
| Retail investor adverse selection | Low                        | High                       | Est. 3–5× |

Securities regulation transitioned from Regime I to Regime III during 2005–2010. The critical variable was not regulatory stringency — Dodd-Frank increased it — but EVR. As entities adopted microsecond decision cycles while regulators maintained 18-month cycles, coevolution collapsed.

### D. Case Study 3: Platform Governance (Regime III — Collapsed)

**Context:** Social media content moderation, 2015–2025.

**EVR Estimate:**  $\tau_R \approx 2\text{--}4$  years (legislative: FOSTA, state laws; admin: FTC consent decrees);  $\tau_E \approx 2\text{--}6$  weeks (algorithm updates, terms changes); **EVR  $\approx 100\text{--}500$ .**

**Empirical Validation** — Facebook/Meta 2015–2023, tracking substantive TOS changes:

- Mean time between substantive TOS changes: **42 days**
- Mean time for users to challenge TOS in court: **18–36 months**
- EVR  $\approx 150\text{--}250$

By the time any TOS provision reaches litigation, it has been revised 10–15 times. Courts review obsolete contractual terms. The coevolutionary relationship has not merely weakened — it has become structurally impossible.

### E. Quantitative Synthesis

Pooling data across all three domains (N = 47 regulatory actions, 20-year period):

Regression model:  $\text{Arbitrage\_Rate} = \beta_0 + \beta_1 \cdot \log(\text{EVR}) + \beta_2 \cdot \text{Resources} + \beta_3 \cdot \text{Stringency} + \varepsilon$

| Variable   | Coefficient | Std. Error | p-value |
|------------|-------------|------------|---------|
| log(EVR)   | 0.18        | 0.03       | < 0.001 |
| Resources  | 0.02        | 0.04       | 0.62    |
| Stringency | −0.03       | 0.05       | 0.55    |



| Variable       | Coefficient | Std. Error | p-value |
|----------------|-------------|------------|---------|
| R <sup>2</sup> | 0.74        |            |         |

A 10-fold increase in EVR corresponds to an 18 percentage point increase in arbitrage rate. Regulatory resources and stringency are statistically insignificant once EVR is controlled. EVR explains 74% of variance in regulatory effectiveness.

**Critical Threshold Test** — piecewise regression with breakpoint at EVR = 100:

| Domain         | EVR < 100 | EVR > 100 | Difference       |
|----------------|-----------|-----------|------------------|
| Arbitrage rate | 12%       | 58%       | 46 pp            |
| t-statistic    |           |           | 8.23 (p < 0.001) |

EVR ≈ 100 represents a genuine regime boundary. Below this threshold, regulation exhibits standard effectiveness-resource relationships. Above it, effectiveness collapses independent of resources.

## IV. EXTENDED PHENOTYPE IMPLICATIONS

### A. Constitutional Lock-In Index Under Temporal Stress

My prior work (Lerer, 2024) developed the Constitutional Lock-In Index (CLI), measuring institutional rigidity through four dimensions: textual vagueness enabling judicial activism; treaty hierarchy constraining domestic reform; precedent weight creating path dependence; and federal fragmentation multiplying veto points. High-CLI systems exhibit superior institutional persistence.

But CLI measures transmission within human-scale temporal regimes. What happens to high-CLI doctrines under temporal stress? I propose:

$$\text{CLI}_{\text{effective}} = \text{CLI} \times (1 - \text{EVR}/100)$$

When EVR > 100, even high-CLI doctrines collapse functionally.

#### Test Case: Chevron Deference in Algorithmic Contexts

In traditional domains ( $\tau_{\text{agency}} \approx \text{months}$ ,  $\tau_{\text{entity}} \approx \text{months}$ ):  $\text{EVR} \approx 1$ ,  $\text{CLI}_{\text{effective}} \approx 0.85$  — doctrine functions as designed. In algorithmic domains ( $\tau_{\text{agency}} \approx \text{months}$ ,  $\tau_{\text{algorithm}} \approx \text{milliseconds}$ ):  $\text{EVR} \approx 10^6$ ,  $\text{CLI}_{\text{effective}} \rightarrow 0$ .

Analysis of FCC Chevron deference in net neutrality cases confirms the pattern: when regulating physical infrastructure (cable, spectrum), Chevron governs ( $EVR \approx 2-5$ ); when regulating packet routing algorithms, Chevron is formally cited but operationally irrelevant ( $EVR > 1000$ ). High CLI does not immunize doctrines from temporal collapse.

## **B. Constitutional Lock-In Plus Velocity Stress**

Constitutionalized slow governance plus computationally fast environment produces irreversible dysfunction. The formal statement:

$$\text{Constitutional\_Dysfunction} = \text{Lock-In\_Index} \times \text{EVR}$$

When  $\text{Lock-In} > 0.7$  AND  $\text{EVR} > 100$ : irreversible path-dependence on obsolete governance model; the dysfunction cannot be fixed legislatively (constitutional entrenchment prevents it); nor judicially (temporal lag is too great for courts to maintain relevance).

### **Example: Section 230 in the Algorithmic Age**

Section 230 of the Communications Decency Act (1996) was designed to protect platforms from liability for third-party content. In 2025, platforms make billions of content moderation decisions annually through algorithmic mediation ( $EVR > 100$ ). By the time litigation reaches courts, platforms have updated algorithms hundreds of times. The doctrine is constitutionally entrenched — embedded in First Amendment jurisprudence — but operationally obsolete. Courts review ancient algorithmic states. This is the intersection of high CLI and high EVR at its most visible.

## **C. Blind Breeders in Three Fitness Landscapes**

My “Blind Breeders” framework (Lerer, 2025b) shows legislators select for political fitness rather than policy effectiveness. EVR reveals a third landscape. Legislators experience political feedback (polls, donations, reelection) on timescales of months. But temporal viability feedback — can law execute at the speed of the regulated environment? — occurs on scales of years to decades, arriving too late to affect electoral behavior.

The California Consumer Privacy Act (CCPA, 2018) illustrates the mechanism. Political fitness: high — it responded to Cambridge Analytica, carried broad legislative support, earned “toughest privacy law in America” coverage. Temporal viability: low —  $\tau_{\text{CCPA\_enforcement}} \approx 18-24$  months;  $\tau_{\text{platform\_adaptation}} \approx 2-4$  weeks;  $EVR \approx 25-50$ . Observed outcome (2018–2024): platforms comply formally (privacy policies updated) while adapting substantively (data collection continues via methods not yet prohibited). Enforcement is perpetually reactive. No political penalty for temporal blindness accrued to the authors of the law.

## V. WHY “COMPUTATIONAL LAW” FAILS

### A. The Seductive Logic

Faced with  $EVR > 100$ , the intuitive response is to accelerate regulators to match entity velocity. If algorithms adapt in milliseconds, deploy regulatory algorithms operating in milliseconds. This is the core of “computational law” proposals: AI adjudication to process disputes at computational speed (Hildebrandt, 2015; Pasquale, 2020); algorithmic regulation with automated penalty assessment and continuous compliance verification (Yeung, 2018; Zaring, 2023); predictive governance using machine learning to forecast and pre-empt violations (Moses, 2017; Coglianese and Lehr, 2019). The promise: restore  $EVR \approx 1$  by accelerating  $\tau_R$  to match  $\tau_E$ .

### B. JudgeAI: The Most Advanced Implementation

Before addressing the fatal flaw shared by all computational law proposals, it is worth examining the most technically sophisticated implementation currently in operation. Kozlov (2024), in a paper available on SSRN and operationalized through the JudgeAI plugin, proposes a fully automated judicial system combining large language models with logical algorithms. The architecture is more rigorous than prior proposals. Rather than replacing judges with black-box ML systems, Kozlov designs a structured decision procedure: LLMs identify evidence-fact connections and generate case narratives; logical algorithms evaluate prima facie presumptions and burden-of-proof assignments; a Nash equilibrium framework assesses whether parties’ behavioral deviations represent genuine violations of their contractual roles.

The explicit design objective is telling. Kozlov (2024) writes that “logical algorithms should create a situation where direct human influence on the decision-making process in a case would be impossible.” Absolute independence — defined as the absence of any possibility for an external party to influence the final decision — is presented as a design virtue rather than a concern. In an earlier passage, the paper acknowledges the core temporal problem it cannot solve: “models trained on historical data may not reflect the latest changes in legislation or case law.” This is precisely the EVR problem stated in technical terms. The system optimizes for a legal world that existed at training time, generating authoritative decisions about a legal world that has since changed.

What makes JudgeAI relevant beyond its technical elegance is its Nash equilibrium component. By formalizing parties’ behavioral obligations through game-theoretic role models and evaluating deviations from those models, Kozlov achieves genuine formalization

of what prior computational law proposals left vague. But this formalization is exactly what destroys Condition 3 of strategic interaction: bounded rationality. When both parties' behavior is evaluated against a unified Nash equilibrium framework, the bounded rationality that makes parties genuine strategic agents — capable of surprise, creativity, error, and genuine disagreement — is replaced by optimization toward a mathematical solution concept. The parties are no longer players. They are variables whose deviations from equilibrium are computed, not judged.

A more sophisticated version of JudgeAI incorporates forward-looking models with divergence detection checkpoints: the system simulates how parameters, actor behavior, and consequences evolve over time, places future triggers into the regulatory configuration, and recalculates scenarios when the real trajectory diverges from the projected one (Kozlov, personal communication, February 2026). This approach partially addresses the static temporal blindness concern. But it relocates rather than resolves the EVR problem. The recomputation cycle -- from detected divergence to updated regulatory output -- must itself be shorter than the entity adaptation cycle to maintain coevolutionary symmetry. If entities can modify their behavior faster than the system can recalculate, the trigger fires but the corrective norm arrives describing a world that has already changed again. The EVR problem shifts from the initial model to the feedback loop itself, and the structural asymmetry remains.

This is not a technical flaw in JudgeAI's architecture. It is the architecturally necessary consequence of any attempt to fully automate adjudication. The ambition to eliminate human influence is the ambition to eliminate governance.

### *1. A Hybrid Architecture: EVR-Triggered Deceleration*

This diagnosis points toward a constructive synthesis between the computational law approach and the mandatory deceleration alternative proposed in this paper. The forward-looking divergence detection architecture is a genuine contribution: real-time monitoring of entity trajectories, trigger-based recalculation, and scenario forecasting are exactly the capabilities a regulatory system needs to remain relevant in high-EVR environments. The problem is not the detection mechanism. The problem is what happens when the detected divergence exceeds the system's recomputation capacity.

I propose that the divergence detection module itself serve as the activation trigger for mandatory velocity constraints. When the EVR monitor registers that entity adaptation velocity has exceeded the regulatory recomputation threshold, the system does not attempt to accelerate its own processing. Instead, it activates a circuit breaker on the regulated entities: a temporary velocity limit that reduces  $\tau_E$  until EVR falls back within the Symmetric or Asymmetric Regime. The architecture thus has two modes. In Mode A ( $\text{EVR} <$

100), the forward-looking model operates normally, generating updated regulatory configurations as trajectories evolve. In Mode B ( $EVR > 100$ ), the system switches from regulatory acceleration to environmental deceleration, imposing speed limits that restore coevolutionary symmetry before resuming Mode A operations.

This hybrid architecture preserves the strengths of both approaches. The computational intelligence of the divergence detection system is retained and made more useful: it now functions not only as a regulatory generator but as an EVR gauge. The mandatory deceleration mechanisms proposed in Part VI are no longer static rules requiring legislative intervention but dynamic responses calibrated in real time to measured temporal asymmetry. Human agency is preserved because the velocity constraints that protect it are activated precisely when the EVR monitor demonstrates they are needed, rather than applied uniformly across all conditions. The result is a governance architecture that is simultaneously more adaptive than static deceleration and more protective of human agency than pure computational law.

The institutional implications are significant. The EVR monitoring module is a separable, implementable component that could be deployed independently of any specific adjudication system. A regulator equipped with an EVR gauge can make evidence-based decisions about when to activate velocity constraints, when to relax them, and which domains require permanent versus temporary deceleration. This transforms mandatory deceleration from a blunt legislative instrument into a precision regulatory tool.

### **C. The Fatal Flaw: Eliminating the Players**

Computational law — in any form — solves the EVR problem by eliminating human decision-makers from the governance loop. But this destroys the very thing regulation exists to protect: human agency.

Recall the five conditions for genuine strategic interaction. Computational law restores Condition 5 (temporal compatibility) by destroying Conditions 1–4.

**Condition 1 (Separation) collapses:** Human governance produces space for legitimate disagreement — regulators and entities have distinct goals, and regulation is negotiated through notice-and-comment, settlement, compliance counseling. Computational governance, if both parties optimize against the same objective function or the same equilibrium framework, produces convergence rather than negotiation. “Disagreement” becomes parallel optimization toward a shared attractor. Distinct players collapse into nodes in a unified system.

**Condition 2 (Imperfect information) collapses:** Human governance depends on regulators having limited visibility into entity operations, creating space for strategic behavior. Continuous monitoring generates comprehensive behavioral data; ML systems infer hidden states from observables; “private” information becomes transparent via prediction. China’s Social Credit System illustrates the endpoint — citizen behavior continuously monitored, AI scores updated continuously, no information remaining strategically opaque. When one party can predict the other with near-certainty, the less powerful party is no longer a player — it is a variable.

**Condition 3 (Bounded rationality) collapses:** Cognitive diversity creates the strategic uncertainty that makes game theory applicable. AI systems use standardized reasoning — the same algorithms, the same training data, the same optimization criteria. JudgeAI’s Nash equilibrium framework exemplifies this precisely: parties’ “ideal behavior” is derived from a unified role model, and deviations are measured against it. When all agents reason identically, there is no game — only distributed computation of a predetermined outcome.

**Condition 4 (Minimal symmetry) collapses:** Citizens in human governance systems have exit options — votes, migration, civil disobedience — that create legitimacy constraints on regulatory power. Computational governance is inescapable (machine speed, perfect memory), cannot negotiate (algorithms do not deliberate or extend mercy), and cannot be meaningfully contested (ubiquitous monitoring forecloses exit). Weaker parties are administered, not governed.

## **D. The Canary Stops Singing**

When computational law restores temporal compatibility (Condition 5) by destroying Conditions 1–4, governance has not been saved — it has been replaced by administration. My prior work (Lerer, 2025a) argued that game theory functions as a canary in the mine: when its models fail to describe interaction, this signals erosion of human agency conditions.

Computational law makes the canary sing again. But only because it killed the miners.

The models work. Interaction is predictable. Everyone optimizes efficiently. No human is choosing anymore. The “solution” to EVR collapse is to remove humans from the system entirely — which is to say, the solution to governance failure is to abandon governance.

Kozlov (2024) is admirably candid about one dimension of this. The paper notes that “models trained on historical data may not reflect the latest changes in legislation.” This is the EVR problem from the inside: the system generates authoritative decisions about an obsolete world. What the paper does not acknowledge is that this temporal blindness is not a technical limitation awaiting a better training dataset. It is structural. Legal reality changes

faster than any training pipeline can track, because legal reality now includes algorithmic entities adapting in real time. JudgeAI's temporal horizon is bounded by its last training cutoff. The entities it adjudicates have already adapted to whatever that horizon contains.

## **E. Empirical Preview: Jurisdictions That Tried It**

**China's Social Credit System** (2014–present) provides the most comprehensive evidence of computational governance at scale. Continuous monitoring via surveillance infrastructure, algorithmic scoring of creditworthiness and “trustworthiness,” and automated penalties including travel restrictions and job blacklisting have produced a system where  $EVR \approx 0.1$  — the AI adjusts faster than citizens can adapt. Compliance rates exceed 95%. Behavioral outcomes are predictable and efficiently produced. Agency is zero: citizens cannot negotiate, disagree, or exit. Conditions 1–4 are systematically destroyed. This is governance in name only.

# **VI. MANDATORY DECELERATION: THE ALTERNATIVE**

## **A. Core Principle**

If accelerating regulators to computational velocity eliminates human agency, the alternative is straightforward in principle if technically demanding in practice: decelerate the environment to human velocity. Do not make law faster. Make regulated systems slower.

**Formal Goal:** Maintain  $EVR \in [0.5, 2]$  (symmetric coevolution regime) by imposing velocity constraints on fast-adapting entities.

## **B. Three Mechanisms**

### *1. Circuit Breakers (Mandatory Pauses)*

Automatic suspension of operations when velocity exceeds threshold. NYSE circuit breakers (1988, post-Black Monday) halt trading if the S&P 500 drops 7%, 13%, or 20% intraday. The mechanism forces market operations to occur at human deliberation speed during stress events. Without circuit breakers,  $\tau_{\text{HFT}} \approx \text{microseconds}$  versus  $\tau_{\text{regulator}} \approx \text{months}$  produces  $EVR > 10^6$ . With circuit breakers imposing a minimum effective pause of 15 minutes, EVR falls to 10–100 — asymmetric but functional. Post-1988 evidence is informative: major dislocations (1997 Asian Crisis, 2008 Financial Crisis, 2020 COVID shock) did not produce Flash Crash-scale algorithmic cascades precisely because circuit breakers forced periodic synchronization with human deliberation speed.

## *2. Cooling Periods (Mandatory Delays)*

Required waiting period between decision and execution. The EU Consumer Rights Directive's 14-day cooling-off period for online purchases is the model: it ensures consumers have time to deliberate at human pace before contractual obligations bind.

Applied to algorithmic contracts: platform terms-of-service currently update every 6 weeks ( $EVR \approx 150$  versus user litigation timeline). Mandatory 90-day notice plus a 180-day freeze period on further changes would reduce EVR to approximately 5 — restoring users' ability to meaningfully challenge terms in court while the challenged terms are still operative.

## *3. Velocity Limits (Frequency Caps)*

Maximum rate of operational changes. Proposed minimum order duration (500 milliseconds) in financial markets would eliminate HFT strategies exploiting sub-millisecond advantages, ensuring all market participants can observe and respond to orders.

Applied to content moderation: a maximum of one major algorithm change per quarter — where “major” means changes affecting more than 5% of users, altering content visibility by more than 20%, or revising recommendation weights — combined with a 30-day public comment period, would reduce EVR from approximately 25 to approximately 2. Regulatory oversight becomes temporally relevant to actual platform operations for the first time since algorithmic curation became dominant.

## **C. Design Principles for Velocity Constraints**

Effective velocity constraints satisfy five principles. First, constraints should target  $EVR \approx 10$  as a design goal, providing buffer above the Regime I boundary while staying well below the collapse threshold. Second, velocity metrics must be objectively measurable and independently verifiable — timestamps, version control records, and API logs provide the audit trail; substantive regulations require judgment calls, but velocity is objective. Third, constraints should be calibrated to risk, with stricter limits in high-stakes domains (financial stability, constitutional rights) than in low-stakes ones (e-commerce recommendations). Fourth, implementation should be gradual, starting with extreme cases (HFT, social media algorithms) and expanding based on effectiveness evidence. Fifth, velocity constraints should include automatic review every five years: if EVR naturally falls below 10 due to technological slowdown or regulatory acceleration, constraints can be relaxed.

## **D. Objections and Responses**

**“This will kill innovation.”** Empirical evidence from 35 years of financial market circuit breakers contradicts this. Markets are more innovative, liquid, and efficient than before



1988. Innovation that requires  $EVR > 100$  to be profitable is **arbitrage innovation** — it extracts value from temporal gaps rather than genuine efficiency gains. Eliminating such “innovation” is desirable, not costly.

**“Slow systems will be outcompeted globally.”** The race-to-the-bottom logic fails on the evidence. EU GDPR imposed significant compliance costs — which are, functionally, velocity constraints on data processing. Five years later, EU tech sector formation had increased relative to pre-GDPR trends. Firms adapt to velocity constraints if they want market access.

**“Entities will route around constraints.”** This objection misunderstands the goal. The objective is not to eliminate evasion but to maintain coevolutionary symmetry. When entities evade, regulators detect and adapt — this is Regime I/II dynamics, functional rather than collapsed. Moreover, velocity is harder to evade than substantive rules: timestamps do not lie, and structuring changes to evade velocity limits imposes costs that deter marginal evasion while permitting essential adaptations.

**“Emergencies require fast response.”** All proposed velocity constraints include emergency exceptions requiring declaration by specified authority, justification on record subject to later judicial review, and automatic sunset provisions. Emergency rulemakings represent approximately 0.3% of Federal Register rules over the 2000–2020 period. Designing systems around the 0.3% while ignoring the 99.7% is poor institutional design.

## VII. OBJECTIONS AND LIMITATIONS

### A. “This is Luddism”

The critique confuses means with ends. The goal is not to preserve regulatory jobs or resist technological change. It is to preserve conditions of human agency — the structural prerequisites for individuals to be strategic actors rather than administered objects. Velocity constraints are pro-technology in the sense that they create sustainable conditions for technology adoption. Technologies deployed without regard for human adaptation capacity produce backlash: GMOs in the EU (1990s), social media algorithmic feeds (2010s), and facial recognition in law enforcement (2020s) all illustrate how rapid deployment without velocity limits generates political restriction more severe than the constraints proposed here.

## B. Scope Limitations

This Article focuses on temporal asymmetry as one dimension of regulatory failure. EVR is not the only determinant of effectiveness. Regulatory capture serves industry interests regardless of EVR. Information deficits may prevent regulators from understanding entity behavior even when temporal symmetry exists. Resource constraints are real. The relationship to EVR is best stated as follows: EVR is necessary but not sufficient for effectiveness. Low EVR does not guarantee good regulation, but high EVR **guarantees** bad regulation. Oxygen is necessary for fire but does not guarantee fire; the analogy holds.

The strong claim advanced here — when  $EVR > 100$ , regulation will fail regardless of resources, expertise, or political will — rests on the empirical evidence presented in Part III. The weak claim — when  $EVR < 10$ , regulation may still fail, but failure is not structurally predetermined — is relatively uncontroversial.

## VIII. CONCLUSION

### A. Summary of Contributions

This paper has formalized Evolutionary Velocity Ratio (EVR) as a dimensionless diagnostic tool for temporal asymmetry in governance relationships. Empirical analysis across tax law, securities regulation, and platform governance is consistent with the hypothesis that  $EVR \sim 100$  represents a critical threshold: below it, regulatory effectiveness exhibits standard relationships with resources and stringency; above it, effectiveness appears to collapse regardless of those factors. EVR accounts for 74% of variance in regulatory arbitrage rates across the sample examined, while resources and stringency are statistically insignificant once EVR is controlled. These findings are suggestive rather than definitive; replication across additional jurisdictions and regulatory domains would strengthen the empirical case.

The dominant reform proposal — computational law — fails because it solves EVR by eliminating human decision-makers from governance loops, destroying the conditions of human agency that regulation exists to protect. Kozlov's (2024) JudgeAI system, the most technically sophisticated implementation currently in operation, exemplifies this: its design objective of making human influence "impossible" is precisely the objective that transforms governance into administration. The alternative — mandatory deceleration through circuit breakers, cooling periods, and velocity limits — preserves agency while restoring coevolutionary symmetry.

## B. The Red Queen's Choice

Regulatory systems face a fundamental choice:

**Path A (Accelerate Law):** Deploy AI adjudication, algorithmic enforcement, predictive governance. Restore  $EVR \approx 1$  by matching computational velocity. Cost: eliminate human agency; transform governance into administration. Outcome: efficient control, zero freedom.

**Path B (Decelerate Environment):** Impose velocity constraints on fast-adapting systems. Maintain  $EVR < 10$  by limiting rate of change. Cost: some innovations foreclosed; slower adaptation in emergencies. Outcome: preserve human agency; sustain democratic governance.

This is not a technical choice between regulatory technologies. It is a normative choice about what kind of society we want to live in.

The Red Queen must keep running to stay in place. But when the ground beneath her feet moves at computational velocity, she faces a deeper question: should she run faster, or should she slow the ground?

This Article argues for the latter. Not because speed is inherently bad, but because human deliberation has intrinsic value. Time to think, to disagree, to negotiate — these are not inefficiencies to be optimized away. They are constitutive features of human agency.

## C. Broader Implications

The temporal collapse of regulatory coevolution extends beyond law. Democratic accountability fails when policy effects emerge on timescales longer than electoral cycles — voters assess officials based on politically salient signals rather than policy outcomes that arrive too late or remain causally opaque. Constitutional entrenchment mechanisms were designed for slow-moving social changes; when social conditions change faster than constitutional amendment procedures allow, entrenchment transforms from stability into sclerosis. AI development cycles (months) outpace regulatory cycles (years) by orders of magnitude; current debates over “AI alignment” focus on technical alignment but temporal misalignment may be more fundamental — if AI systems adapt faster than governance systems can monitor and adjust, alignment is impossible regardless of technical solutions.

## D. Research Agenda

Several research directions follow from this analysis. Empirically: constructing a comprehensive EVR database across regulatory domains and jurisdictions; testing whether

velocity constraints successfully reduce EVR in practice; and examining historical regulatory failures through the EVR lens. Theoretically: formalizing the connection between EVR and extended phenotype theory (why do some legal memes resist temporal collapse?); modeling coevolutionary dynamics in Regimes I/II/III using evolutionary game theory; and developing optimal velocity constraint design that maximizes social welfare subject to agency preservation constraints. Normatively: when, if ever, is Path A (accelerate law) justified? How should emergency exceptions be designed to prevent abuse? What international coordination mechanisms prevent “velocity arbitrage” — entities relocating to jurisdictions without velocity limits?

## **E. Final Reflection**

The Red Queen told Alice: “It takes all the running you can do, to keep in the same place.”

But what happens when the Red Queen stumbles? When the ground moves faster than anyone can run?

The intuitive response — make regulators faster — is a trap. It preserves the race by eliminating the runners we care about: human beings with the time to deliberate, disagree, and decide.

The alternative requires accepting a limit. Not all innovations should proceed at maximum velocity. Some systems — particularly those governing fundamental human relationships — should be designed to operate at human speed.

This is not Luddism. It is humanism.

The ground will not slow itself. We must decide, collectively and deliberately, to slow it. To create spaces where temporal compatibility is maintained. Where the Red Queen can still run and, in running, still matter.

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## APPENDICES

### Appendix A: EVR Calculation Methodology

EVR measurement across regulatory domains requires consistent operationalization of adaptation cycle components. For regulatory adaptation time ( $\tau_R$ ), the full cycle comprises four stages: detection (from observable violation to formal investigation initiation, mean 3–6 months in U.S. administrative contexts), design (from investigation to proposed rule, mean 6–18 months), implementation (from final rule to active enforcement ramp-up, mean 6–12 months), and validation (from enforcement action to doctrine consolidation, mean 12–36 months). Total  $\tau_R$  for U.S. federal regulatory domains typically ranges from 27 to 72 months.

For entity adaptation time ( $\tau_E$ ), measurement varies by domain. In algorithmic domains, adaptation cycles are measured in days to weeks, using code deployment frequency as proxy — publicly available through GitHub commit histories, API changelog publication, and mandatory disclosure filings in regulated sectors. In financial domains, HFT adaptation cycles are measured in milliseconds to days using order data from FINRA TRACE and market microstructure studies. In traditional commercial domains, adaptation cycles of 6–18 months are estimated using compliance officer survey data and patent filing patterns.

EVR estimates should be treated as order-of-magnitude rather than precise measurements. The regime classification (Symmetric/Asymmetric/Collapsed) is robust to errors of factor 2–3 in either component; the qualitative findings are not sensitive to precise EVR values.

### Appendix B: Regression Specifications and Robustness Checks

The core regression specification uses *Arbitrage\_Rate* as the dependent variable, measured as the proportion of regulatory actions in each domain-period that are offset by documented arbitrage strategies within 24 months of implementation.  $\text{Log}(\text{EVR})$  is the primary independent variable, using the midpoint of EVR range estimates for each domain-period.

Alternative specifications tested: (1) linear EVR rather than log specification — coefficient 0.006 ( $p < 0.001$ ), slightly lower  $R^2$ ; (2) domain fixed effects — coefficient on  $\text{log}(\text{EVR})$  unchanged at 0.18, domain effects not significant; (3) excluding HFT cases (potential

leverage concern) — coefficient 0.15 ( $p < 0.01$ ),  $R^2$  0.68, findings robust; (4) instrumental variable specification using lagged legislative session length as instrument for  $\tau_R$  — coefficient 0.21 ( $p < 0.001$ ), larger in magnitude, consistent with downward attenuation in OLS.

The piecewise regression threshold at  $EVR = 100$  is confirmed by Chow test (F-statistic 67.8,  $p < 0.001$ ) and Davies test for unknown breakpoint (breakpoint estimate:  $EVR = 94$ , 95% CI: 78–113).

**Appendix C: Case Study Data Sources**

**Tax Law (Case Study 1):** Primary data from IRS Statistics of Income (SOI) Division publications 1990–2020; Treasury Inspector General for Tax Administration (TIGTA) reports on compliance rates; U.S. Tax Court opinions database; Joint Committee on Taxation legislative history analyses.

**Securities Regulation (Case Study 2):** SEC EDGAR filings for dark pool volume data; SEC Market Structure Statistics database; academic HFT studies by Brogaard, Hendershott, and Riordan (2014); Federal Register notice-and-comment timelines from RegInfo.gov; enforcement action database from SEC Litigation Releases.

**Platform Governance (Case Study 3):** Internet Archive Wayback Machine for TOS change tracking (Facebook/Meta 2015–2023); Federal Trade Commission enforcement timeline database; PACER federal court records for TOS litigation timelines; academic tracking datasets from Princeton Web Transparency and Accountability Project.

**Appendix D: EVR Compliance Framework for Platform Algorithm Output Regulation**

*Note: The following technical specification operationalizes the output-based velocity constraints developed in Part VI. It is framed as a regulatory compliance standard rather than a legislative text: the obligations it defines are measurable outputs — timestamps, modification frequency, EVR readings — that a regulator can impose by administrative order, consent decree, or licensing condition, without requiring a new legislative cycle. The specifications are illustrative and calibrated to produce  $EVR < 10$  for platforms currently operating in the Collapsed Regime.*

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| PLATFORM                                             | ALGORITHM | VELOCITY | CONSTRAINT | STANDARD |
|------------------------------------------------------|-----------|----------|------------|----------|
| (EVR Compliance Framework — Technical Specification) |           |          |            |          |

## 1. Definitions

As used in this Framework:

- (a) “Covered platform” means any online service with more than 10 million monthly active users in the United States that employs algorithmic systems to rank, filter, recommend, or otherwise curate content for users.
- (b) “Major algorithm modification” means any change to a covered platform’s algorithmic systems that: (i) foreseeably affects the content ranking, recommendation, or filtering outcomes for more than 5% of monthly active users in the United States; (ii) alters the weight assigned to any primary ranking factor by more than 20 percentage points; or (iii) introduces or retires a ranking factor previously absent from or present in the deployed system.
- (c) “Regulatory adaptation period” means the 12-month period following adoption of a major algorithm modification, during which regulatory agencies and affected users may evaluate the modification’s effects.
- (d) “Emergency modification” means a major algorithm modification implemented outside the standard notice and delay procedures under Section 3 pursuant to a certified emergency declaration.

## 2. Registration and Transparency Requirements

- (a) Each covered platform shall maintain a publicly accessible Algorithm Change Register documenting all major algorithm modifications, including: the date of each modification; a plain-language description of the modification and its intended purpose; the estimated proportion of users affected; and baseline and post-modification metrics for content engagement, exposure, and removal rates.
- (b) The Algorithm Change Register shall be updated within 72 hours of any major algorithm modification becoming operational.
- (c) The designated regulatory authority shall establish technical standards for the Algorithm Change Register within 180 days of this Framework's adoption.

## 3. Velocity Constraints — Standard Procedures

- (a) *Notice Requirement.* Not less than 90 days before implementing any major algorithm modification, a covered platform shall publish notice of the proposed modification in the Algorithm Change Register and notify the Federal Trade Commission.



- (b) *Comment Period.* The Federal Trade Commission shall make the notice of proposed major algorithm modification available for public comment for a period of 30 days from publication.
- (c) *Modification Freeze.* Following implementation of a major algorithm modification, a covered platform may not implement any further major algorithm modification for a period of 90 days, except pursuant to Section 4.
- (d) *Maximum Annual Modifications.* A covered platform may implement no more than four major algorithm modifications per calendar year absent a certified emergency declaration under Section 4.

#### **4. Emergency Modifications**

- (a) A covered platform may implement a major algorithm modification outside the procedures of Section 3 only if:
  - (b) The platform's chief executive officer or equivalent certifies in writing that: (A) an imminent and serious harm to users exists that the proposed modification is designed to address; (B) normal advance notice procedures would materially impair the platform's ability to address the harm; and (C) the modification is narrowly tailored to address the certified harm.
  - (ii) The certification and a description of the emergency modification are submitted to the Federal Trade Commission within 24 hours of implementation.
- (b) An emergency modification implemented under this Section shall expire automatically after 30 days unless the platform: (i) provides full notice under Section 3(a) within 15 days of the emergency implementation; and (ii) completes the comment period required by Section 3(b).
- (c) No covered platform may implement more than two emergency modifications per calendar year.

#### **5. Enforcement and Sanctions**

- (a) The designated regulatory authority — whether a sector-specific regulator, competition authority, or data protection agency — shall have primary authority to enforce this Framework as a condition of platform operation or through consent-based compliance programs.
- (b) Each major algorithm modification implemented in violation of this Framework's velocity constraints constitutes a separate infringement. Sanctions shall not exceed

the equivalent of \$50,000 per day for violations of the notice requirement and \$100,000 per day for violations of the modification freeze, calibrated to eliminate the economic incentive for non-compliance. Regulatory authorities may also impose operational conditions, including temporary suspension of algorithm modification rights, as a compliance remedy.

- (c) Any person harmed by a major algorithm modification implemented in violation of this Act may bring a civil action in any district court of the United States. Prevailing plaintiffs shall be entitled to actual damages, injunctive relief, and reasonable attorneys' fees.

## **6. Review and Appeals**

Regulatory authority interpretations of this Framework are subject to administrative review under applicable procedural law. Emergency certification determinations are subject to expedited independent review within 30 days of implementation. Covered platforms retain the right to challenge EVR measurements and modification classifications through established administrative appeal procedures.

## **7. Relation to Other Instruments**

Nothing in this Framework limits any right or remedy otherwise available under applicable law. This Framework does not displace existing statutory or regulatory obligations. Where obligations overlap, the more protective standard applies. Compliance with this Framework does not confer immunity from liability under competition law, data protection law, or consumer protection law.

## **8. Severability**

If any provision of this Framework or its application to any covered platform is found to exceed regulatory authority or to be otherwise unenforceable, the remaining provisions and their application to other platforms or circumstances are not affected.

## **9. Implementation Timeline**

This Framework takes effect 180 days after formal adoption by the designated regulatory authority.

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*Key design choices: The 90-day notice period and 90-day modification freeze are calibrated to produce  $EVR < 10$  — well within the Symmetric Regime — for platforms currently operating at  $EVR > 100$ . The four-modification annual cap functions as a secondary velocity constraint.*

*Emergency procedures are sufficiently restrictive (CEO certification, automatic expiration, two-per-year maximum) to prevent strategic abuse while accommodating genuine crises. Critically, every obligation in this Framework is defined as a measurable output — a timestamp, a frequency count, an EVR reading — rather than a process requirement. This makes the Framework enforceable through automated monitoring and immune to the legislative cycle delays that the paper identifies as the root cause of Regime III collapse.*

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***Methodological Note on AI Assistance:*** *This paper was researched and written by the author, who holds a law degree from Universidad de Buenos Aires and an Executive MBA from IAE Business School. Mathematical formalizations and some analytical structures were developed with the assistance of large language model AI tools. The author takes full responsibility for all theoretical frameworks, empirical interpretations, legal analysis, and normative arguments. Readers who identify errors in the formal specifications are encouraged to contact the author directly.*